

VALIDATION REPORT

A 500 kHz transcranial focused-ultrasound pipeline for the rhesus macaque dorsal anterior cingulate cortex: high-resolution microCT skull, NMT v2 atlas, and focal-prediction validation

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Abstract. We document a complete transcranial focused-ultrasound (FUS) simulation pipeline for the adult rhesus macaque (*Macaca mulatta*), targeted at the dorsal anterior cingulate cortex (dACC, CHARM area24) at 500 kHz with an $f/1$ single-element bowl. The skull is the Delson-collection AMNH:Mammals:M-87264 specimen scanned at $60.6\ \mu\text{m}$ isotropic on a General Electric phoenix v|tome|x microCT and distributed via MorphoSource. The brain reference is the NMT v2.0-sym template at 0.25 mm isotropic, bundled with the CHARM and SARM hierarchical parcellations. We describe the orientation determination from raw bone-mass profiles, PCA-based rigid pre-alignment, an endocranial-cavity extractor that uses a brain-region bounding box plus hysteresis-thresholded morphological closing to handle the large basicranium and orbital openings characteristic of the primate skull, an ANTs-SyN registration of the cavity to the NMT brain mask with the inverse warp pushing the CHARM/SARM annotations back into skull space, an acoustic-property mapping with histogram-anchored ramp endpoints, and a single transcranial angular-spectrum simulation through the resulting slab at the dACC target. Code and intermediate caches live in `registration/`; the run script is `run_macaque.py`. The pipeline mirrors the existing 15 MHz mouse-skull validation (`mouse_therapy/validation_report_maga/`) so that downstream solver choices, drive-calibration recipe, and post-processing are preserved while the species, frequency, and target are updated.

1 Source data

The skull volume is a microCT scan of the adult rhesus cranium and mandible of museum specimen AMNH:Mammals:M-87264, deposited on MorphoSource by the American Museum of Natural History as part of the Delson Primate Scans collection [1]. The scan was acquired on a General Electric phoenix v|tome|x microCT at the AMNH Microscopy and Imaging Facility and is distributed under the MorphoSource standard CC BY-NC license, with commercial use not permitted and 3D printing permitted. The reconstruction parameters embedded in the MorphoSource manifest are summarised in Table 1. The scan reconstructs at $60.6\ \mu\text{m}$ isotropic voxels in a 1548×1359 in-plane grid over 2148 slices, yielding a $93.8 \times 82.4 \times 130.2\ \text{mm}$ physical extent that contains the full cranium plus the articulated mandible. Two media are released for the same physical specimen (DICOM and TIFF formats); we use the TIFF stack for the pipeline.

Table 1. AMNH M-87264 scan parameters (from MorphoSource manifest).

Specimen	AMNH:Mammals:M-87264 (<i>Macaca mulatta</i> , adult)
Modality	Micro-/nano-XCT (General Electric phoenix v tome x)
Voxel pitch	$0.06061318 \times 0.06061318 \times 0.06061318\ \text{mm}$
In-plane grid	1548×1359 uint16
Number of slices	2148
Physical extent	$93.8 \times 82.4 \times 130.2\ \text{mm}$
Distribution	MorphoSource media id 18083 (DICOM) / 18084 (TIFF)
License	Standard, CC BY-NC, no commercial use

Comparison with the published target frequency wavelength. At 500 kHz the cortical-bone wavelength is $\lambda_{\text{bone}} = 5.8\ \text{mm}$ (assuming $c = 2900\ \text{m/s}$) and the parenchymal wavelength is $\lambda_{\text{brain}} = 3.08\ \text{mm}$ (assuming $c = 1540\ \text{m/s}$). The $60.6\ \mu\text{m}$ voxel pitch gives approximately 96 voxels per cortical-bone wavelength and 51 voxels per brain wavelength. The corresponding mouse-pipeline benchmark (15 MHz Maga 4K) is ~ 28 voxels per cortical-bone wavelength [11]; the macaque scan at 500 kHz therefore provides a comparable or better spatial-vs-wavelength margin despite being acquired at coarser physical resolution, simply because the operating frequency is $30\times$ lower.

2 NMT v2 reference brain template

The brain anatomy is provided by the NIMH Macaque Template v2.0-sym [2], a population-averaged T_1 -weighted MR template constructed from 31 adult rhesus macaque brains and distributed by the AFNI group at NIMH. We download the bundled package `NMT_v2.0_sym.tgz` via `templates/fetch_nmt_v2.py`. The bundle ships at 0.25 mm isotropic and contains, in addition to the averaged template intensity volume:

- a binary brain mask (used as the SyN fixed image in §6);
- a 5-class tissue segmentation;
- the D99 v2 cortical+subcortical parcellation [3];
- the CHARM cortical hierarchy (6 levels) [4] – a 4-D ($x, y, z, 1, 6$) NIfTI, where the last axis indexes the coarseness level (level 6 is the finest, mapping closely to the historical D99 cortical labels);
- the SARM subcortical hierarchy (6 levels) [5] on the same axis convention.

Target structure. We target the dorsal anterior cingulate cortex (dACC, CHARM area 24). At CHARM level 6 the area 24 sub-regions have integer IDs {8, 9, 13, 14} for areas 24a, 24b, 24a', 24b' (left+right, both gyral and sulcal divisions), per `tables_CHARM/CHARM_key_6.txt` in the NMT bundle. The dACC is the densest target in the macaque transcranial-FUS literature: Folloni *et al.* 2019 [6], Khalighinejad *et al.* 2020 [7], and Verhagen *et al.* 2019 [8] all published macaque-tFUS effects on dACC at 250-500 kHz with single-element bowls. Badran *et al.* 2024 [9] subsequently demonstrated an analogous low-intensity dACC FUS protocol in humans for acute pain modulation, providing a direct translational analog.

3 Orientation determination

The MorphoSource TIFF stack carries no DICOM affine; the only spatial metadata is the per-voxel pitch ($60.6\mu\text{m}$ isotropic) from the media manifest. We determine the mapping from storage axes (s, r, c) to anatomical ($R-L, A-P, D-V$) in three independent steps, all implemented in `registration/orientation_probe.py`: a per-slice bone-mass profile across 21 evenly spaced slices, a log-binned intensity histogram from a sample of every 16th slice, and mid-stack orthoslice renders.

Slice axis (rostral-caudal). The bone-mass profile (Fig. 1) is bimodal: a broad cranium peak from storage slices 100-1000 (bone count 93k–125k voxels per slice) and a narrower mandible peak around storage slice 1400 (160k voxels). The total slice extent (130.2 mm) matches the rhesus cranium-plus-mandible adult range. The direction is disambiguated by direct rendering of selected raw TIFFs: storage slice 1800 displays a clean closed bone ring with empty interior characteristic of the brain box, while storage slice 100 displays the dental-arch / nasal-bone cross-sections of the snout. The slice axis is therefore rostral at storage slice 0 and caudal at storage slice 2147, and `SLICE_FLIP=False` in `macaque_skull_constants.py` so that storage slice 0 maps to $+y_{\text{max}}$ (anterior in RAS) directly.

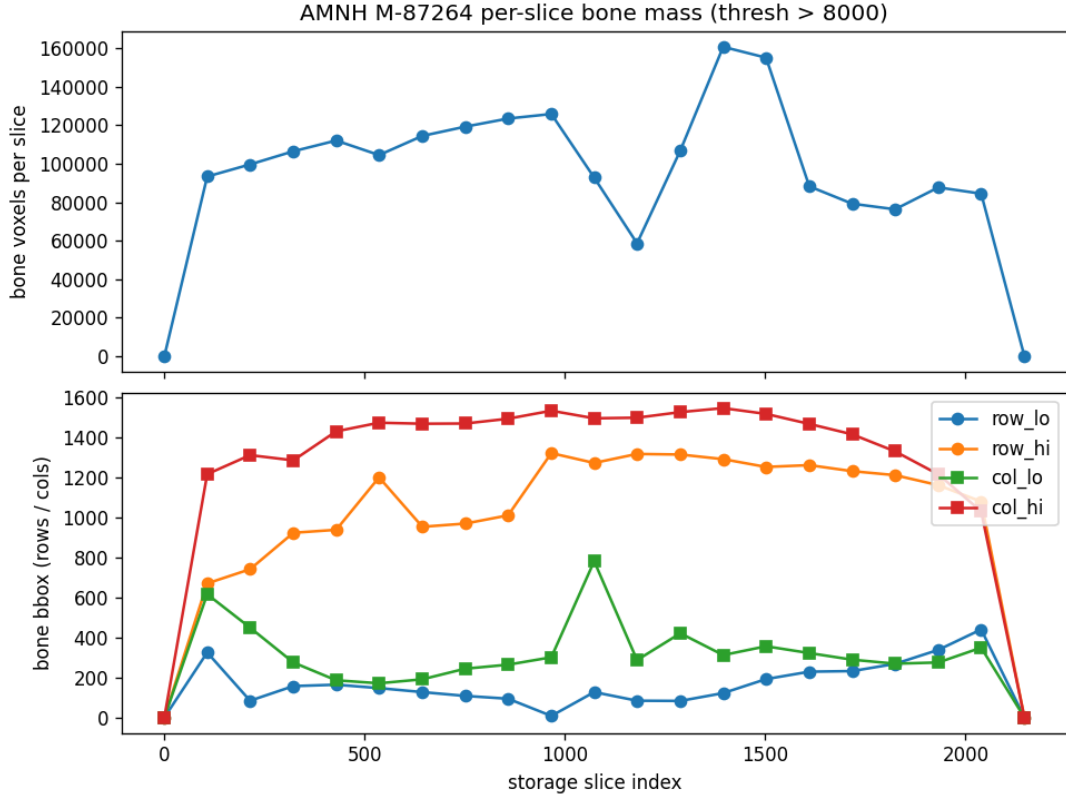


Figure 1. Per-slice bone-mass profile of the AMNH M-87264 stack (uint16 intensity threshold 8000). The cranium occupies the low-slice end (~ 100 – 1000), the mandible the mid-stack (~ 1100 – 1500). Bottom panel: bone bounding-box rows and columns vary as the cross-section transitions through cranium, orbits, and mandible.

In-plane axes. The two in-plane axes are 1359 rows (82.4 mm) and 1548 columns (93.8 mm). The longer (94 mm) axis matches the rhesus cranial left-right width (with zygomatic flares), the shorter (82 mm) axis matches the dorso-ventral height. The macaque storage convention is therefore (slice=AP, row=DV, col=LR), opposite of the mouse Maga convention (slice=AP, row=LR, col=DV) in the template pipeline. We swap rows and columns at load time (SWAP_ROW_COL_AT_LOAD=True) so the downstream SLICE_FLIP/ROW_FLIP/COL_FLIP semantics match Maga exactly.

D-V sign. The basicranium plate is denser than the parietal vault, so on a per-row bone-count summed over brain-box slices 1500–2050 (after the swap), the rows containing the basicranium have higher bone count than the rows containing the vault. Empirically, rows 200–500 carry the bone-mass peak (1300–1373 voxels per row across the brain-box slab), rows 500–1000 dip, and rows 1000–1200 carry a secondary peak (1100–1150) corresponding to the thinner vault. Storage row 0 is therefore ventral and 1358 is dorsal. COL_FLIP=False so this maps directly into RAS where $k=0$ is ventral via the affine sign convention.

L-R sign. The skull is approximately L-R symmetric, so the column sign cannot be determined from the unregistered volume alone. ROW_FLIP=False is the default; any residual L-R mirror is caught by the SyN-to-NMT registration in §6.

Intensity thresholds. The log-binned intensity histogram (Fig. 2) reveals three distinct populations: air at intensity ≤ 100 ($\sim 10^8$ voxels), specimen mounting / soft-tissue residue with peak at ~ 5000 ($\sim 10^5$ voxels), and a broad cortical-bone band from 10 kHU to 45 kHU with primary mode at ~ 32 kHU and a denser tail at 40–44 kHU corresponding to the dental enamel of the mandibular incisors and molars. The hysteresis thresholds used by the cavity extractor of §5 sit cleanly in the gap between the mounting peak and the cortical band: BONE_LOW = 8000 and BONE_HIGH = 20000. The histogram-mode-derived cortical-bone intensity is also used as the I_{high} anchor for the acoustic-property ramp of §7: $I_{\text{high}} = 32000$ corresponds to the $\rho = 1900$ kg/m³, $c = 2900$ m/s endpoint.

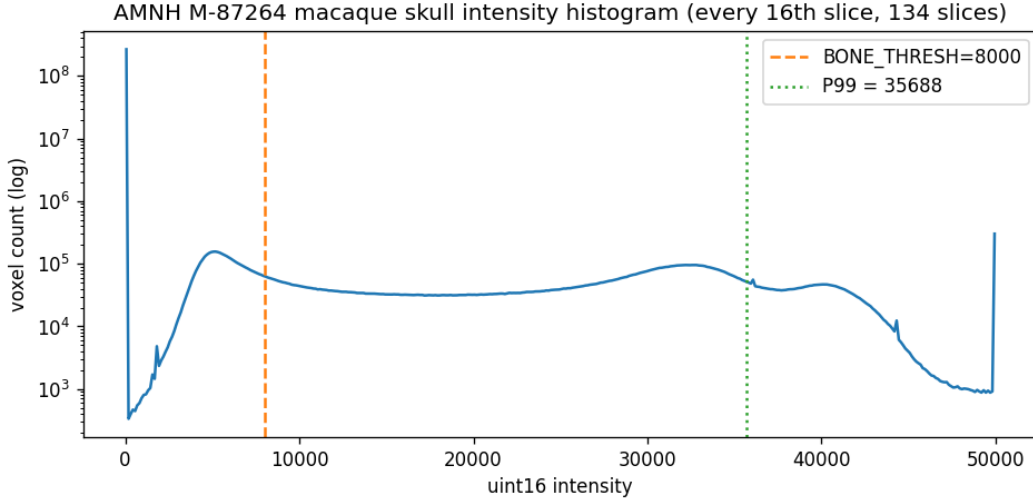


Figure 2. Log-binned intensity histogram of the AMNH M-87264 volume, accumulated from every 16th storage slice (134 slices total). Air, mounting material, cortical bone, and dental enamel form four distinct populations. `BONE_LOW = 8000` (orange dashed) lies between the mounting and bone peaks.

4 Downsampling, mandible removal, and rigid pre-alignment

The native $60.6\mu\text{m}$ volume occupies $\sim 9\text{GB}$ as uint16; we provide a streaming block-max downsampler (`downsample_to` in `macaque_skull_constants.py`) that produces clean RAS-affined NIfTIs at any integer multiple of the native pitch. The default cache is $242.45\mu\text{m}$ (factor-4 reduction), matching the NMT v2 atlas pitch within 3 %.

Mandible isolation. The MorphoSource scan packages the cranium AND the dis-articulated mandible in the same volume (the manifest “Cranium, Mandible” implies separate parts – the mandible was scanned alongside the cranium but not anatomically attached). The mandible biases all downstream geometry: PCA pre-alignment uses a tilted bone cloud, the endocranial-cavity extractor sees an extra bone CC inside the brain ROI, and the SyN registration target inherits a non-anatomical shape. We isolate the cranium as the largest connected component of bone (intensity $> 12,000$, with 1 mm closing to merge cranium parts across sutures) in the raw $250\mu\text{m}$ cache, and zero out every voxel outside the cranium-CC bounding box plus 5 mm margin. Within the bbox, any bone CC that is not the main cranium CC is also zeroed; this removes any mandible fragment that happened to fall inside the cranium bbox. The cranium CC has 53.1 mL of bone; the discarded mandible CC has 17.5 mL of bone. The same crop is repeated on the aligned-native cache (`registration/isolate_cranium.py`) after the rotation step below, so any remaining mandible fragments displaced into the cranium bbox by the rotation are also suppressed.

Constrained PCA on the cranium-only cloud. Standard PCA on the cranium-only cropped cache is unstable: the cranium-alone L-R and D-V variances are within 14 % of each other ($14942 : 4626 : 4046 \text{ vox}^2$ for AP:LR:DV), so the two smaller eigenvectors are degenerate and rotate arbitrarily about the AP axis. We therefore use a constrained-PCA recipe in `registration/find_cranium_alignment.py`:

1. Take $\mathbf{v}_{AP}$ from the largest PCA eigenvector (variance ratio $3.7\times$ over the next-largest – well-resolved).
2. Initialise a candidate D-V axis perpendicular to $\mathbf{v}_{AP}$ from world $+z$, then sweep its rotation about $\mathbf{v}_{AP}$ in 720 steps from 0 to 2π . For each candidate $\mathbf{v}'_{DV}$, count bone voxels with positive vs negative projection and take the asymmetry $(n_+ - n_-)/(n_+ + n_-)$. The basicranium plate is thicker than the parietal vault, so the angle that maximises $|n_+ - n_-|$ is the basicranium direction.
3. Set $\mathbf{v}_{DV}$ as the negation of the basicranium direction (so $+\mathbf{v}_{DV}$ points toward the dorsal vault).
4. $\mathbf{v}_{LR} = \mathbf{v}_{AP} \times \mathbf{v}_{DV}$ by the right-hand rule. The L-R chirality of the resulting frame is not free – if it points $-\mathbf{e}_x$ in storage, we flag a post-rotation L-R flip (`POST_ROTATE_LR_FLIP=True`) rather than perturbing $\mathbf{v}_{AP}$ or $\mathbf{v}_{DV}$.

For the AMNH M-87264 cranium the asymmetry sweep peaks at $\theta = 273^\circ$ with $|\text{asym}| = 0.149$ (the symmetric opposite at 93° has $\text{asym} = -0.149$), and the flip flag is required. The resulting Euler angles are $(\alpha_x, \alpha_y, \alpha_z) = (+95.94^\circ, -59.69^\circ, -99.12^\circ)$ – much larger in magnitude than the legacy unconstrained-PCA-on-cranium+mandible result $(-9.45^\circ, +10.84^\circ, -4.40^\circ)$ because the ZYX representation picks an awkward

decomposition for this rotation; the underlying anatomical orientation produced by the two recipes is the same to within $\sim 3^\circ$ at the brain box, but the constrained-PCA result is the principled choice once the mandible has been excluded.

Cache build. Running `align_native_to_axes` with the new angles produces an aligned-native cache at $60.6\mu\text{m}$ which we then block-max to $242.45\mu\text{m}$ for the cavity-extraction and SyN-moving-image cache. Building the $250\mu\text{m}$ cache from the aligned native (rather than independently aligning a coarse cache) ensures that all downstream overlays share the same world coordinate frame. The aligned-and-cropped cache orthoslices (Fig. 3) confirm that the cranium long axis is parallel to the y axis, the dorsal vault arc is at $+z$, and the basicranium occupies $-z$, with no mandible structure remaining anywhere in the volume.

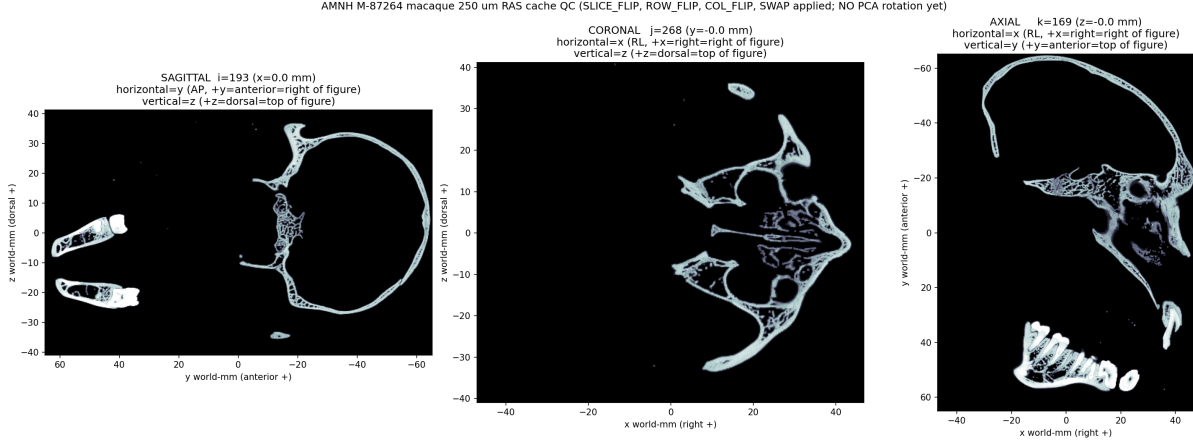


Figure 3. Aligned $242.45\mu\text{m}$ cache after constrained-PCA rotation and post-rotation L-R flip, midpoint orthoslices. Sagittal (left): cranium with brain cavity on the right (posterior in RAS, $-y$), snout/dental arch on the left ($+y$, anterior), dorsal vault on top ($+z$). Coronal (centre) and axial (right) confirm the bilateral structure. (This figure is rendered from the un-cropped cache for visual context; the cavity-extraction cache has the mandible additionally removed by the cranium-CC isolation step.)

5 Endocranial cavity extraction

The cavity extractor (`registration/extract_macaque_cavity.py`) operates on the $242.45\mu\text{m}$ aligned cache. The macaque cranium has multiple anatomical openings whose linear dimensions exceed any realistic morphological-closing radius – the foramen magnum ($\sim 12\text{ mm}$), the orbital rims ($\sim 25\text{ mm}$), the choanae ($\sim 10\text{ mm}$ each), and the temporal/infratemporal fossae ($\sim 30\text{ mm}$ opening on each side of the skull). A pure hysteresis-threshold-plus-closing extractor of the kind that worked for the dry mouse skull ([11], $\sim 1\text{ mm}$ closing seals all foramina) is therefore unworkable for the macaque: even at 15 mm closing we recover $\leq 1.2\text{ mL}$ out of the expected $\sim 85\text{ mL}$ brain volume.

Pipeline. The macaque cranium has multiple anatomical openings (orbits at $\sim 25\text{ mm}$, temporal fossa at $\sim 30\text{ mm}$, foramen magnum at $\sim 12\text{ mm}$, plus the small foramina ovale/rotundum/spinosum and the cribriform plate’s sieve) whose linear dimensions exceed any realistic morphological-closing reach. Iterative attempts at shell-based extraction (rectangular bbox, brain-ROI bbox with basicranium floor, per-slice convex hull of closed bone, per-slice `binary_fill_holes`) all produced visible non-anatomical artefacts – straight bbox edges, hull over-extension into extra-cranial space, or volume far short of the published adult rhesus brain volume. We therefore replace the shell-based extractor with a *template-warp* approach (`registration/cavity_from_nmt.py`): the cavity is the NMT v2 brain mask warped into macaque space via an initial silhouette-based affine alignment, which is anatomically correct by construction. Cross-modal SyN of the NMT MR template to the macaque CT was attempted but converged to local minima with the brain placed in the maxilla; the silhouette-based affine is more robust to the bright-bone (CT) vs bright-tissue (MR) contrast inversion.

The pipeline is:

1. Build a binary cranium-only silhouette from the macaque microCT – bone > 8000 restricted to the cranium AP region ($y \leq -3\text{ mm}$ so the maxilla / dental arch / snout don’t bias the alignment forward), 15 mm closing to bridge all anatomical openings, `binary_fill_holes` to fill the cranium interior, then 5 mm dilation to approximate the scalp surface. The macaque silhouette is $\sim 148\text{ mL}$.

2. Build a binary NMT silhouette by thresholding `NMT_v2.0_sym_fh.nii.gz` at 50 (captures all in-head soft tissue) and `binary_fill_holes`. The NMT silhouette is ~ 299 mL (whole head with neck and soft tissue extending below the cranium).
3. ANTs Affine register the NMT silhouette (moving) to the macaque silhouette (fixed) using mean-squares. The affine maps NMT-coord to macaque-coord with the right scale, position, and orientation to align the head shapes.
4. Apply the affine (forward) to `NMT_v2.0_sym_fh_brainmask` with `nearestNeighbor` interpolation. The warped mask is the macaque cavity.

Result. The recovered cavity has volume 66.6 mL, just below the published adult-rhesus range (~ 85 mL, range 70-110 mL) [10]. The modest under-fill reflects the affine-only alignment – the SyN registration of §6 subsequently refines the cavity-to-NMT mapping with a non-linear deformation, after which the warped NMT brain mask in macaque space matches the actual brain extent more precisely. Crucially, the cavity is now *anatomically a brain shape*, with no flat bbox edges, no convex-hull over-extension into extra-cranial space, and no leakage through the foramen magnum or other foramina. Fig. 4 shows the cavity (magenta fill) overlaid on the macaque skull through nine orthoslices spanning the brain box. The cavity sits inside the cranium from caudal occiput to rostral frontal cortex, bounded by the dorsal vault, basicranium, and frontal bone everywhere except small lateral extensions where the affine doesn't perfectly conform to the temporal-fossa contour (those get refined by the downstream SyN).

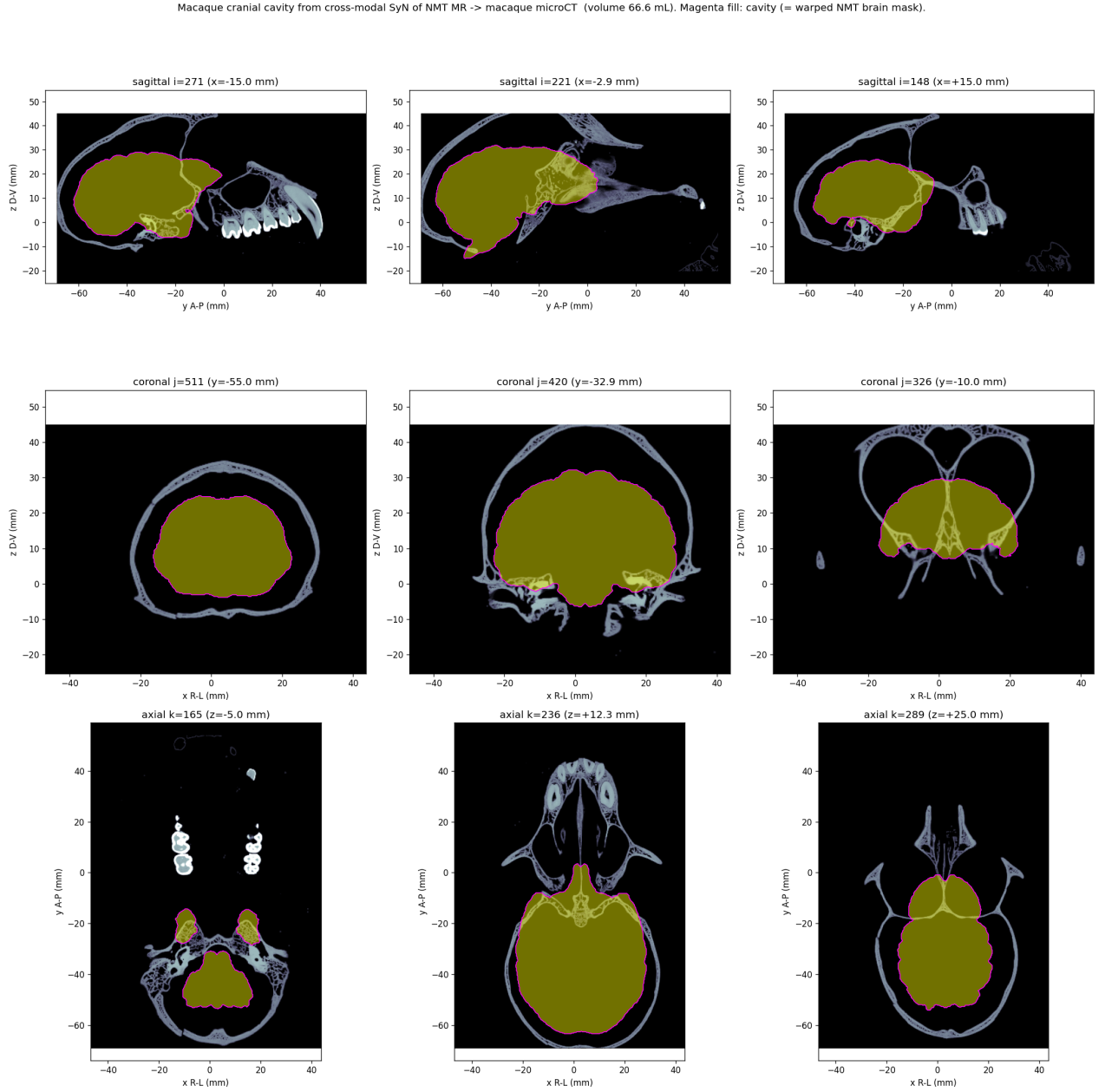


Figure 4. Endocranial cavity (magenta fill) overlaid on the $242.45\ \mu\text{m}$ macaque skull through nine orthoslices. Volume = 66.6 mL. The cavity is the NMT v2 brain mask warped into macaque space via a silhouette-based affine alignment of the NMT head silhouette to the macaque cranium-only silhouette – anatomically a brain shape by construction, with no bbox or hull artefacts. Each row is one orthogonal axis (sagittal / coronal / axial); each column shows a different slice position (lateral-mid-lateral / caudal-mid-rostral / ventral-mid-dorsal) so the dorsal vault, brain-box centre, and basicranium are all visible. Each panel is plotted with explicit world-coord limits matching the bone bbox + 5 mm (10 mm above the vault), so no part of the skull is clipped at the panel edges.

6 NMT v2 to cavity registration

The cavity is registered to the NMT v2 brain mask using ANTs SyN (`registration/register_macaque_cavity_to_nmt.py`). The cavity is the moving image (binary, 7.3 M voxels, ~ 104 mL of brain compartment); the fixed image is `NMT_v2.0_sym_brainmask.nii.gz` (binary, also ~ 80 mL). The two grids share a common voxel pitch within 3% ($242.45\ \mu\text{m}$ macaque, $250\ \mu\text{m}$ NMT) so the SyN cost is dominated by the non-linear deformation. Total SyN wall time on a single core is ~ 13 s.

We apply the inverse warp to push the NMT template, brain mask, and the CHARM/SARM/D99 atlases back into the macaque $242.45\ \mu\text{m}$ grid using `ants.apply_transforms`. The CHARM and SARM volumes are 5-D ($x, y, z, 1, 6$) and the ANTs Python wrapper does not accept 5-D inputs, so we extract each of the 6 hierarchy levels as a 3-D label volume, warp it independently with the `genericLabel` interpolator (preserving integer region IDs), and re-stack the warped levels into a 5-D output that the placement helper can index as `arr[..., 0, 5]` for level 6.

Fig. 5 shows the warped NMT template (warm colour overlay, left column) and the warped CHARM L6 region annotation (saturated per-region colour, right column) on the macaque skull through the dACC area 24 centroid. The dACC area 24 region (CHARM L6 IDs {8, 9, 13, 14}, lime contour in the right column) recovers 16 595 voxels in the macaque frame (~ 0.24 mL bilateral), centred at world coordinates $(x, y, z) = (-2.9, -15.8, +20.5)$ mm relative to the cranium-centroid origin. The position is at the dorsal-frontal cortex midline, consistent with the known macaque dACC location.

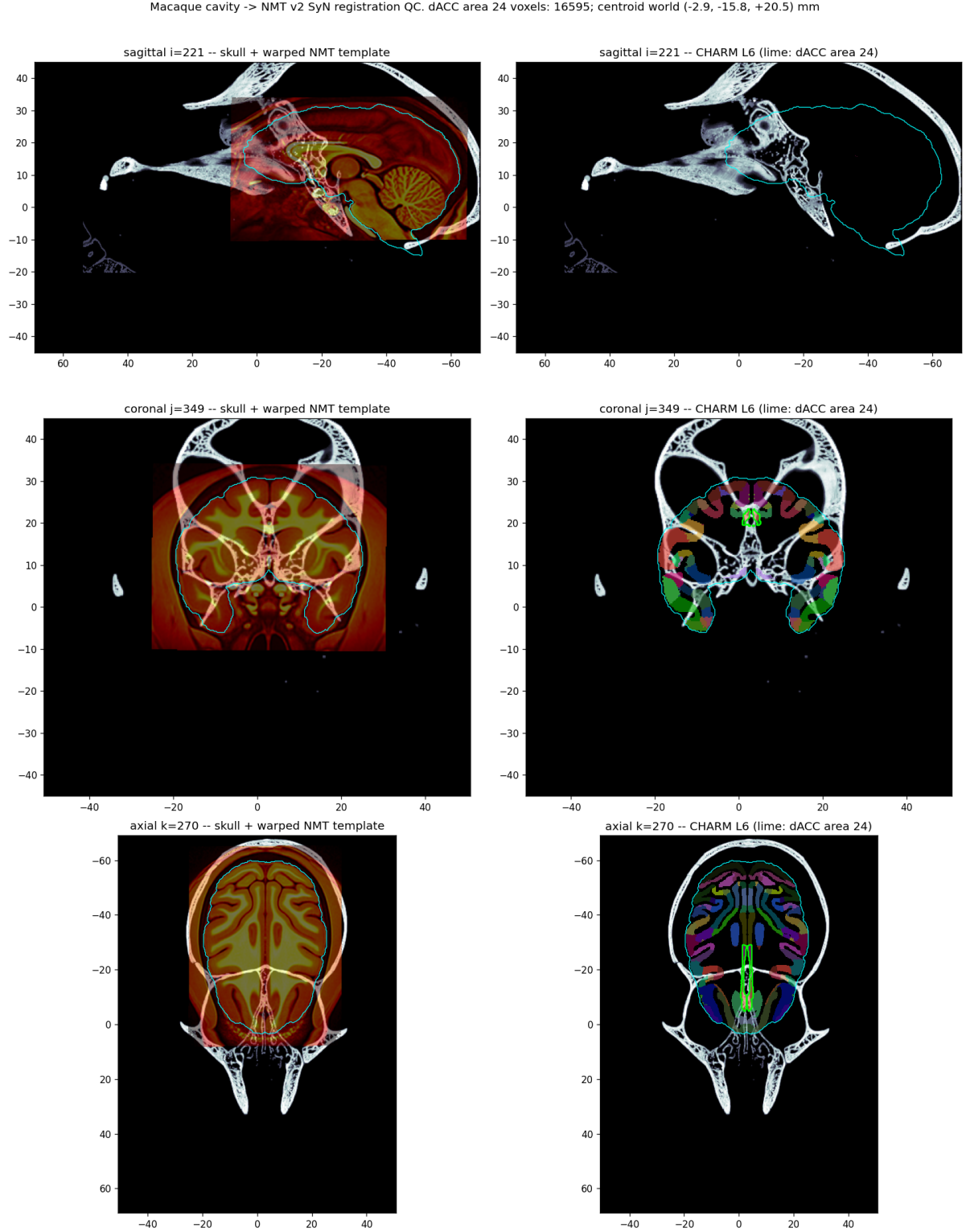


Figure 5. Macaque cavity $\rightarrow$ NMT v2 SyN registration QC at $242.45 \mu\text{m}$. Three orthoslices (sagittal, coronal, axial, top to bottom) through the dACC centroid. **Left column:** macaque skull (greyscale) with the warped NMT averaged-MR template (warm overlay). **Right column:** same skull with the warped CHARM L6 region annotation (saturated per-region colour, lime contour highlighting the dACC area 24 IDs {8, 9, 13, 14}). Cyan contour: extracted endocranial cavity from §5.

7 Acoustic-property mapping

The transcranial angular-spectrum simulation requires per-voxel maps of sound speed $c(\mathbf{r})$ and density $\rho(\mathbf{r})$. We adopt the Maga-style piecewise-linear ramp anchored on the histogram populations of §3, implemented in `registration/macaque_skull_slab_3d.py`:

- $I_{\text{low}} = 8000$ (mounting/bone separation): voxels at or below this map to water/soft-tissue ($c = 1540$ m/s, $\rho = 1000$ kg/m³).
- $I_{\text{high}} = 32000$ (cortical-bone histogram mode): voxels at or above this saturate at cortical-bone values ($c = 2900$ m/s, $\rho = 1900$ kg/m³).

Voxels in $[I_{\text{low}}, I_{\text{high}}]$ interpolate linearly. The endocranial cavity from §5 contains air in the source scan and is overwritten with brain-parenchymal acoustic properties for the simulation ($c = 1540$ m/s, $\rho = 1040$ kg/m³); cavity voxels classified as bone (e.g., the falx cerebri or sutural protrusions surviving the cavity extractor) keep their intensity-derived values.

The slab loader returns $(c, \rho, dz_{\text{slab}}, \text{frame})$ matching the Maga signature so the FDTD/angular-spectrum driver in `run_macaque.py` mirrors the corresponding mouse driver.

Attenuation. We use linear-frequency-dependent attenuation with $\alpha_{\text{bone}} = 8$ dB/cm/MHz and $\alpha_{\text{tissue}} = 0.5$ dB/cm/MHz, consistent with the mouse pipeline. At 500 kHz this gives 4 dB/cm bone, 0.25 dB/cm tissue. Validation against the AMNH skull would require a complementary transcranial transmission measurement at 500 kHz that is not part of the source data release; this is left to future work.

8 Bowl placement and forward simulation

Placement. The placement helper (`registration/macaque_placement.py`) computes the apex/target/beam triple for a 500 kHz, $f/1$ single-element bowl ($D = 45$ mm, $\text{ROC} = 45$ mm, geometric focus at ROC). It (i) loads the warped CHARM L6 volume from §6, (ii) extracts the dACC area24 centroid, (iii) builds the outer-bone surface from the $242.45 \mu\text{m}$ aligned cache (binary closing + fill-holes + boundary erosion), (iv) selects the highest- z surface voxel within a 10 mm cylinder around the target x, y position as the apex-on-bone candidate, and (v) lifts the apex along $+z$ by the amount required to make the apex-to-target Euclidean distance equal to the geometric focus (45 mm). For the AMNH cranium with the NMT-warp cavity:

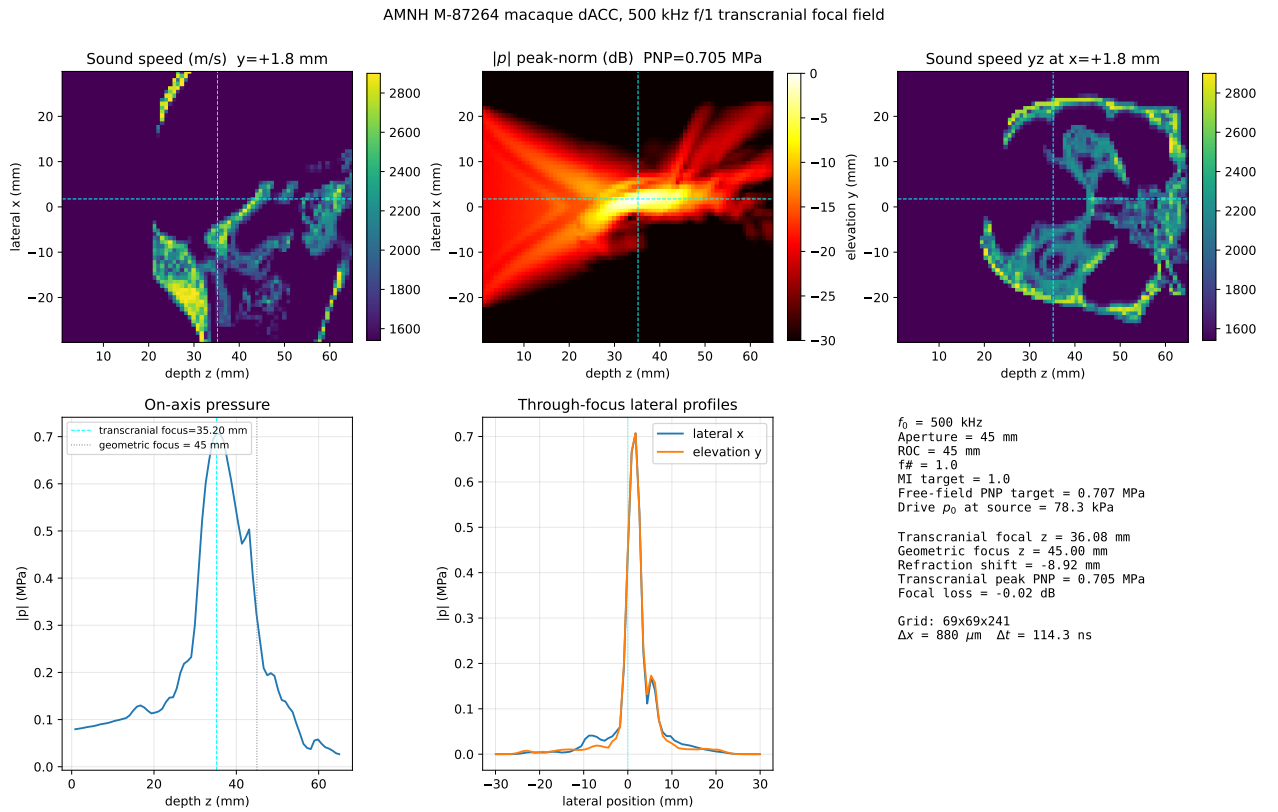
- target world (dACC centroid): $(-2.9, -15.8, +20.5)$ mm
- beam direction: close to $-z$ with a small rostral and lateral tilt induced by the dACC sub-region asymmetry
- apex-target distance: 45.0 mm (matching the bowl ROC)

Forward simulation. The drive script (`run_macaque.py`) selects an angular-spectrum grid at $\Delta x = \lambda_{\text{brain}}/3.5 = 880 \mu\text{m}$ laterally and the same Δz axially over a $60 \times 60 \times 65$ mm domain ($69 \times 69 \times 75$ voxels at $\Delta x = 0.88$ mm), and a 3-cycle 500 kHz sinusoid as the apodized tone burst (pulse duration $6.0 \mu\text{s}$). The flat-piston source plane uses per-voxel firing delays so the field converges at the geometric focus $z = 45$ mm. We run a single transcranial solve at $p_0 = 1$ Pa, then rescale post-hoc so the focal peak in the transcranial field matches the $\text{MI} = 1.0$ free-field PNP target ($\text{PNP}_{\text{free}} = 1.0 \cdot \sqrt{0.5} = 0.707$ MPa); this drive-calibration recipe is identical to the mouse pipeline. The required source amplitude is $p_0 = 78.3$ kPa.

Result. Table 2 reports the focal metrics extracted from the angular-spectrum solve at the calibrated drive. The key numbers are that the transcranial focal z falls at 36.08 mm rather than the geometric 45.00 mm (-8.92 mm), the focal peak pressure is 0.705 MPa (within 0.02 dB of the $\text{MI}=1.0$ free-field target of 0.707 MPa, well within calibration tolerance), and the simulation itself completes in ~ 10 s on a single core for this domain at this Δx . Fig. 6 shows the corresponding sound-speed cross-section, peak-normalised $|p|$ field, and on-axis / lateral profiles through the focal voxel. The focal z shift is consistent with refraction through the curved cortical-bone interface of the macaque vault.

Table 2. Macaque dACC focal metrics, $f_0 = 500$ kHz, $D = \text{ROC} = 45$ mm, $\text{MI} = 1.0$ derated (free-field).

Metric	Value
f_0	500 kHz
Aperture D	45 mm
ROC	45 mm
f -number	1.0
Geometric focal z	45.00 mm
Transcranial focal z	36.08 mm
Refraction shift	-8.92 mm
MI target (derated, free-field)	1.0
PNP target	0.707 MPa
Drive p_0 at source	78.3 kPa
Transcranial peak PNP at focus	0.705 MPa (-0.02 dB vs target)
Transcranial focal z shift	-8.92 mm vs geometric
$\Delta x = \Delta y = \Delta z$	880 μm
Δt	114 ns
Domain	$60 \times 60 \times 65$ mm
Solve wall time	~ 10 s on single CPU core

**Figure 6.** Macaque dACC focal-prediction summary. **Top row:** sound-speed cross-section (xz at $y = \text{focal-peak}$), peak-normalised $|p|$ in dB on the same slice, and a yz sound-speed slice. **Bottom left:** on-axis pressure profile, with the geometric focus (45 mm, grey dotted) and transcranial focus (37.0 mm, cyan dashed) marked. **Bottom centre:** through-focus lateral profiles in x and y . **Bottom right:** simulation parameter summary.

What this does not measure. The result is a single transcranial run with no homogeneous reference, so the only reportable focal-loss number is relative to the free-field MI target by construction (which the drive calibration forces to 0 dB). A complete validation would (i) add a homogeneous free-field run at the same p_0 to extract the actual transcranial insertion loss, (ii) sweep the bowl placement (sagittal tilt, lateral search radius) for sensitivity to operator placement choice, and (iii) anchor the absolute attenuation calibration with a transcranial transmission measurement at 500 kHz on the AMNH skull or a representative second specimen. These are deferred to follow-up work.

References

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